Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 25-03-2026

Relational Inventory Control &Stock Tracking System

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

Presented By:

J.Venkata Sai Charan – VTU26442 –

Reg.No.23UECS0230

Courses Handling Faculty:

Dr.M.Mageshwari

INTRODUCTION

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- The **Relational Inventory Control & Stock Tracking System** is designed to efficiently manage and monitor inventory in a structured and automated way.
- Traditional inventory systems are often manual, time-consuming, and prone to errors in stock calculation and record maintenance.
- Manual tracking of stock levels, purchases, and sales can lead to inaccuracies, stock shortages, or overstocking issues.
- There is a need for a reliable digital system that can handle inventory operations quickly and accurately.
- The Relational Inventory Control & Stock Tracking System addresses these challenges using database-driven technology.
- This project is important because it reduces manual effort, improves accuracy, and ensures real-time stock updates.
- It is highly useful in real-world applications such as warehouses, retail stores, supermarkets, and supply chain management.

PROBLEM STATEMENT

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Relational Inventory Control & Stock Tracking System:

- In many businesses and organizations, inventory management is still handled manually or with basic systems.
- Manual stock tracking consumes time and increases the chances of human error in maintaining records.
- Managing product details, stock levels, purchases, and sales requires significant effort and can lead to inaccuracies.
- Traditional systems lack real-time updates, making it difficult to track available stock efficiently.
- Handling large amounts of inventory data becomes complex and inefficient without proper organization.
- There is a need for a secure and automated inventory management system.
- The system should efficiently track stock, update inventory in real-time, and generate accurate reports.
- Therefore, a Relational Inventory Control & Stock Tracking System is required to improve efficiency, accuracy, and data management.

OBJECTIVE(S)

- To design and develop a Relational Inventory Control & Stock Tracking System that enables efficient management of products, stock levels, and transactions.
- To implement an automated system that updates inventory in real-time during purchases and sales.
- To maintain a structured relational database for storing and retrieving inventory data accurately.
- To generate reports such as stock availability, sales records, and purchase history for better decision-making.
- To reduce manual effort, minimize errors, and improve overall efficiency in inventory management.

PROPOSED SYSTEM

- The proposed system is a web-based Relational Inventory Control & Stock Tracking System with secure user access.
- It allows users to manage product details, stock levels, purchases, and sales digitally.
- The system automatically updates inventory in real-time whenever stock is added or removed.
- It eliminates manual record-keeping, reducing time consumption and human errors.
- The system provides alerts for low stock levels and helps in efficient inventory planning.
- It improves efficiency by handling large amounts of inventory data in an organized relational database.
- The solution ensures accuracy, faster processing, and easy maintenance of inventory records.

PROPOSED SYSTEM ARCHITECTURE

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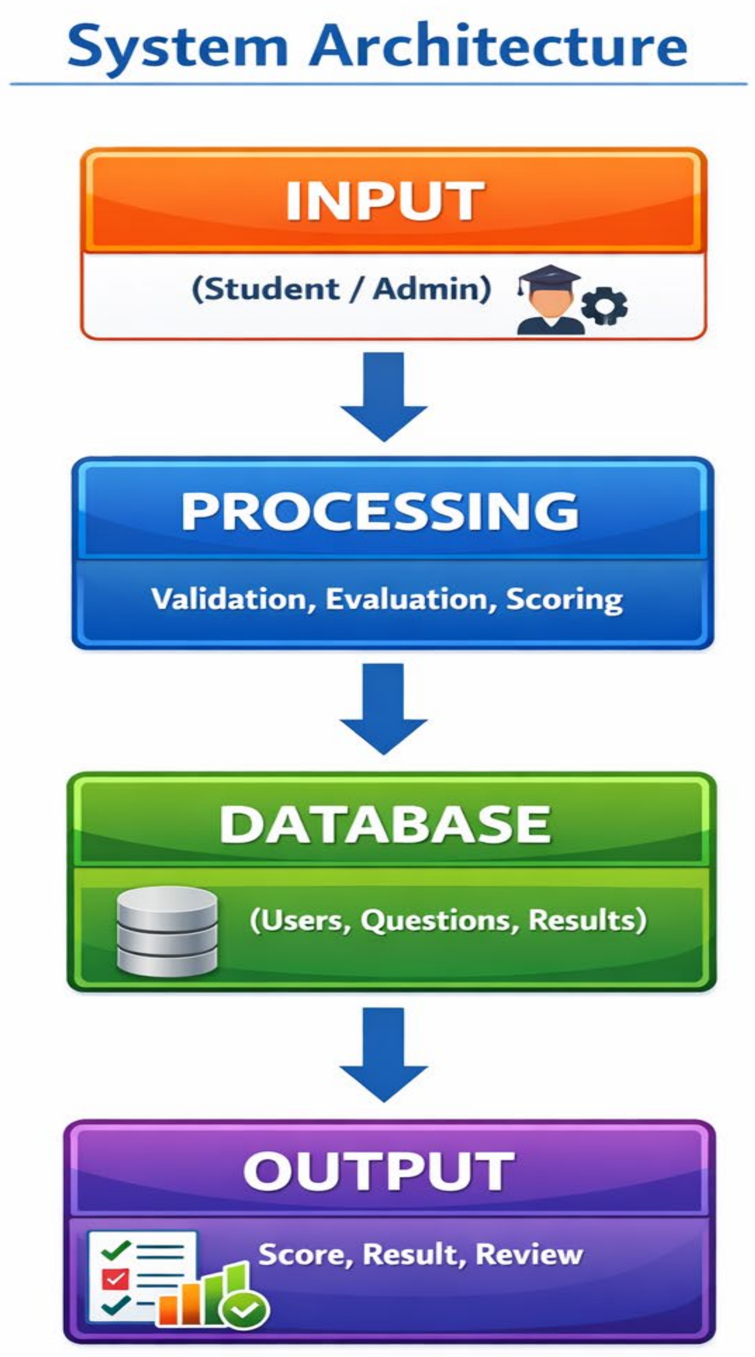

WORKFLOW

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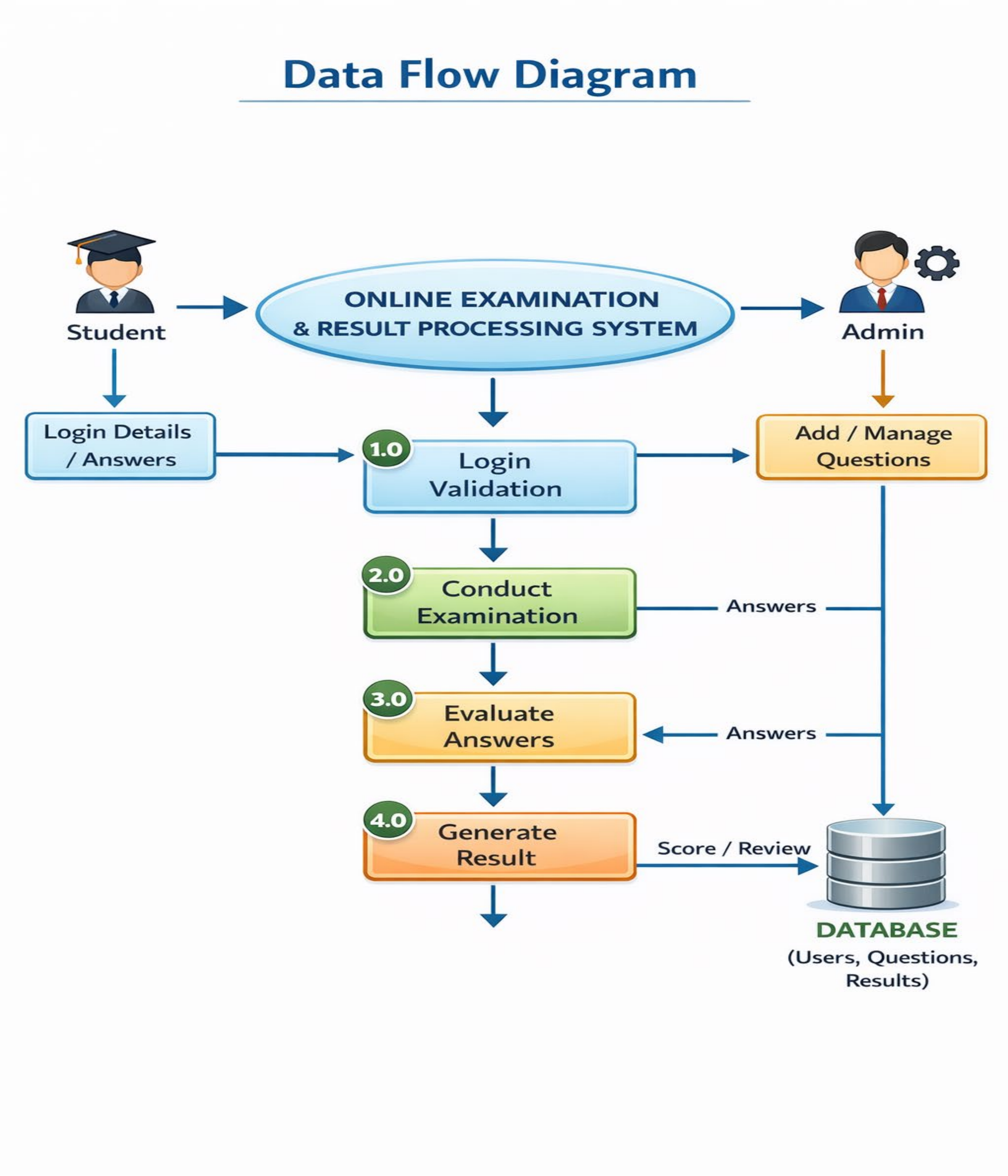


TECHNOLOGIES USED

Frontend:
HTML
CSS
JavaScript
Backend:
JavaScript (for login validation and automatic result evaluation)
(Advanced Version: Python – Flask / PHP)
Database:
Browser-based Storage (Arrays / Local Storage)
(Advanced Version: MySQL / SQLite)
Tools Used:
Notepad
Web Browser (Google Chrome)
Hardware Requirements:
Computer / Laptop
Minimum 4 GB RAM
Internet Connection (for online deployment version)
Software Requirements:
Windows, Web Browser
Text Editor (Notepad)

MODULE DESCRIPTION

Module 1: User Authentication & Login Management
Handles secure student login by validating username and password. Controls access to the examination system and ensures only authorized users can take the test.

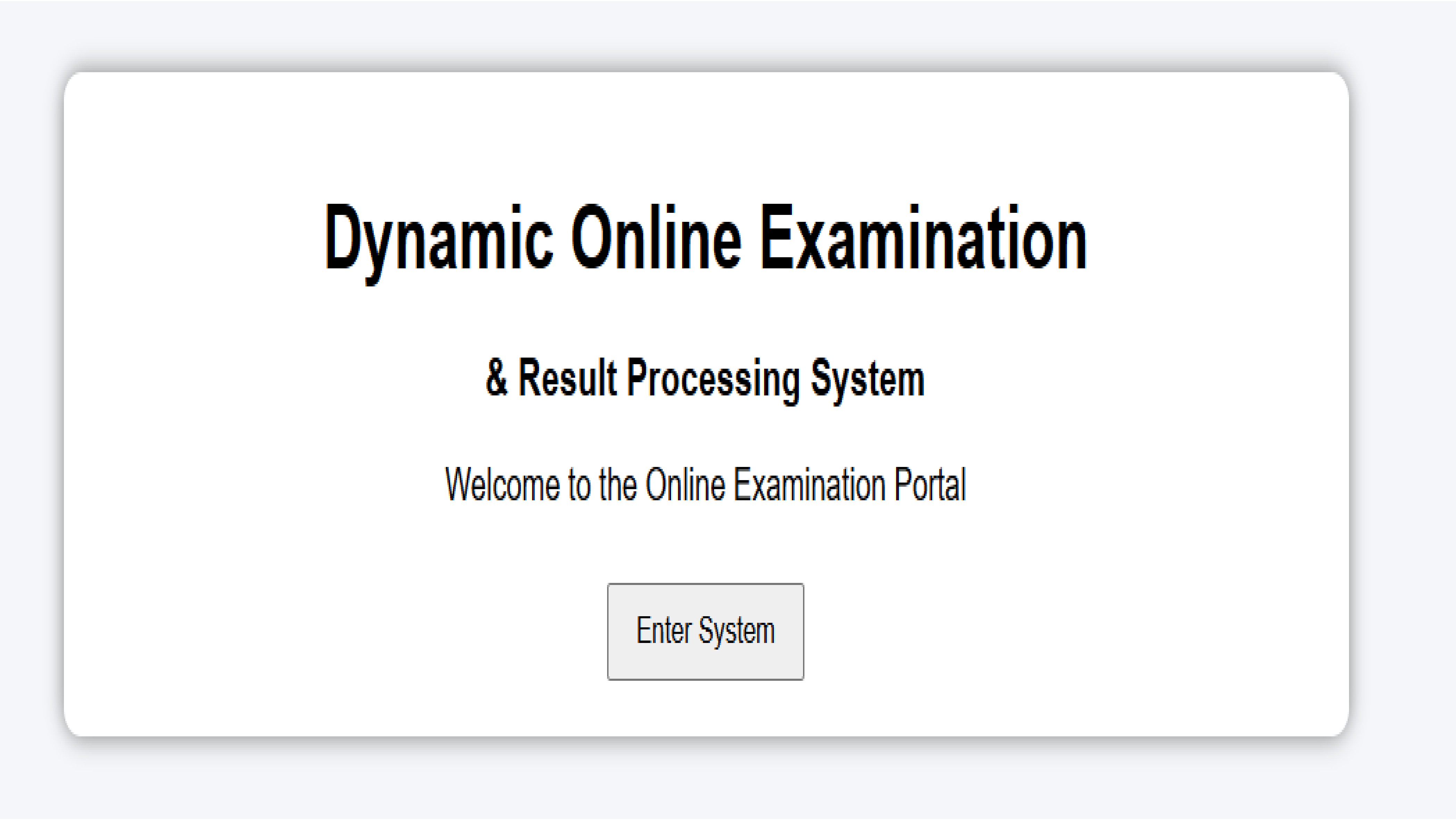
Module 2: Examination Management
Displays multiple-choice questions dynamically and allows students to select and submit answers within the system interface.

Module 3: Evaluation & Result Processing
Automatically compares student answers with correct answers, calculates total score, identifies wrong/unattempted questions, and generates instant results.

Module 4: Data Management & Storage
Manages question bank, user credentials, and exam results using structured data storage to ensure accuracy and consistency.

IMPLEMENTATION SCREENSHOTS

Home page: The Home Page serves as the welcome interface of the Dynamic Online Examination & Result Processing System. It displays the project title and provides an entry point to access the system. This page introduces users to the platform and ensures a simple navigation flow by directing them to the login page. It creates a user-friendly first impression of the system.



IMPLEMENTATION SCREENSHOTS

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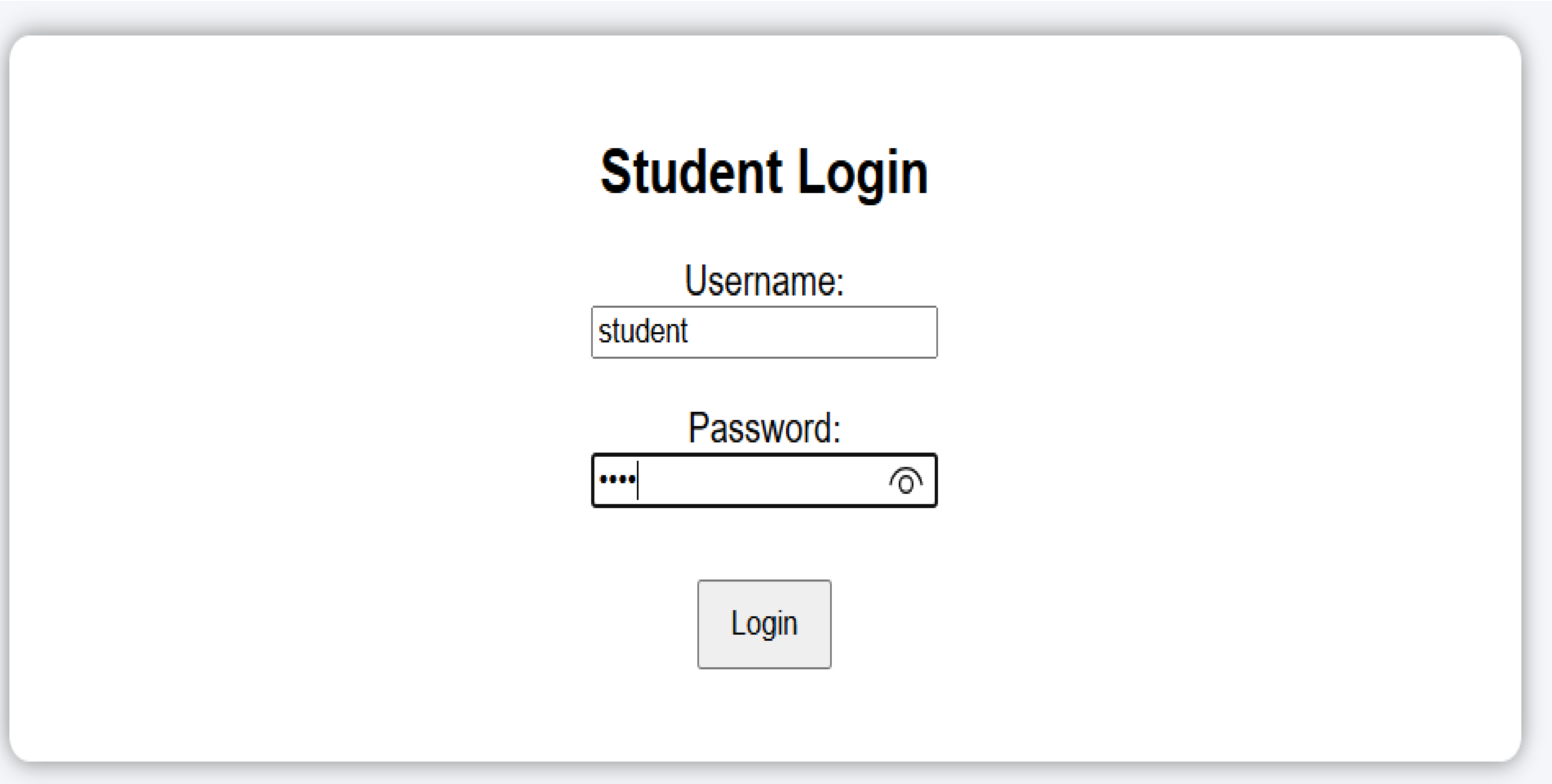
Login page: The Login Page allows students to securely access the examination system using valid credentials. It verifies the username and password before granting access to the exam. This ensures that only authorized users can participate in the examination. It acts as a basic authentication and security module of the system.



IMPLEMENTATION SCREENSHOTS

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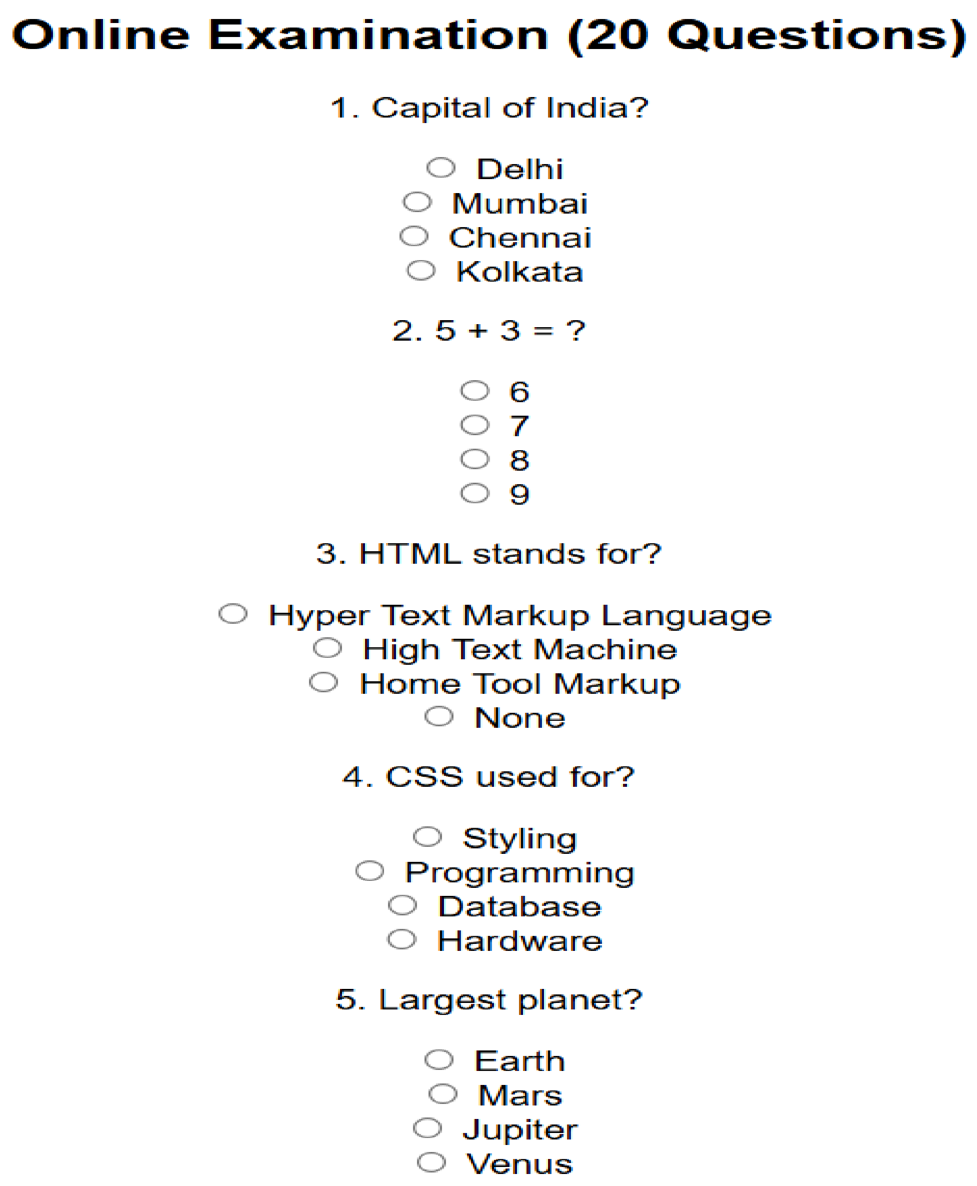
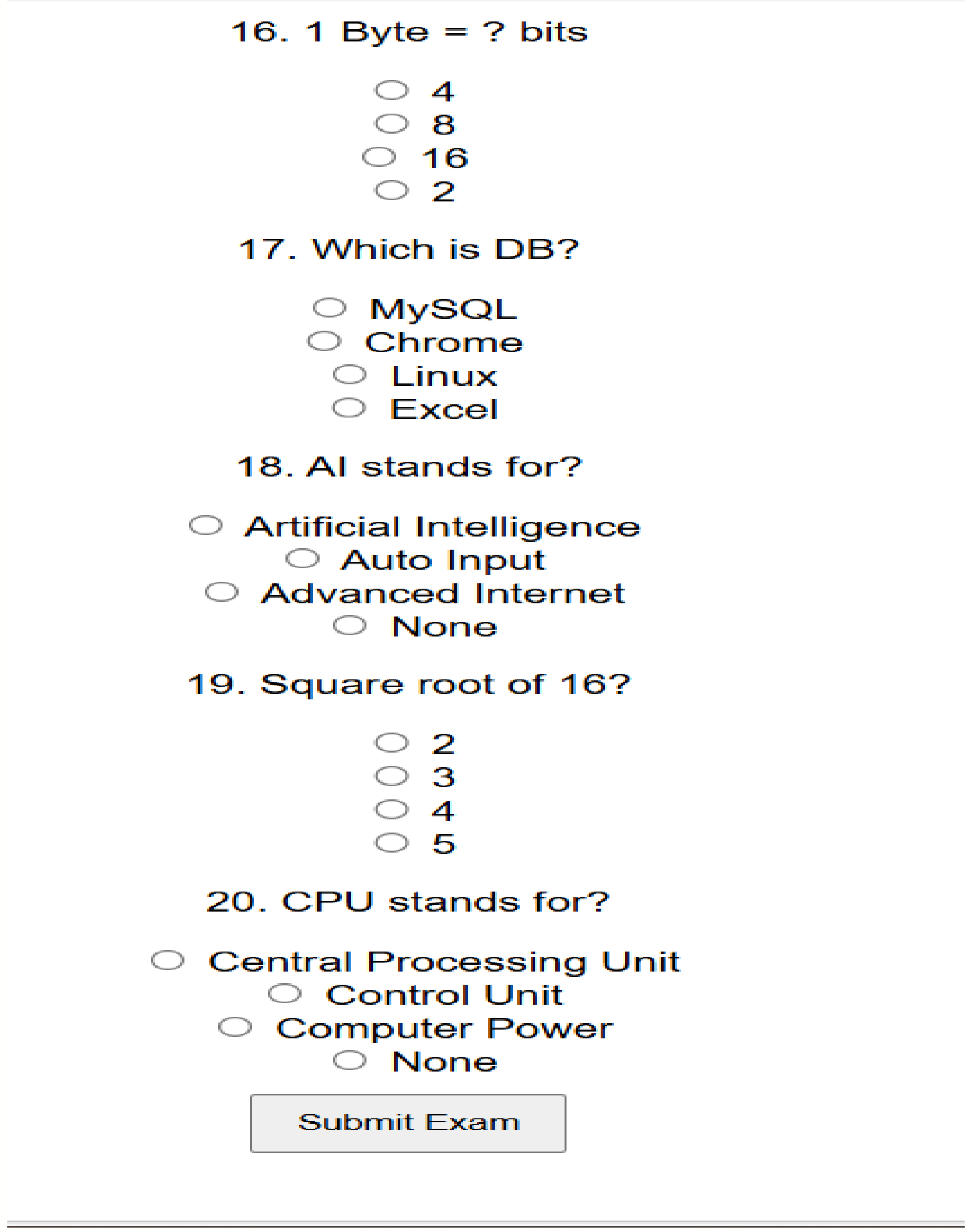
Main module: The Examination Page displays multiple-choice questions dynamically to the student. Users can select answers and submit the exam after completion. The system collects all selected responses and prepares them for automatic evaluation. This module handles the core functionality of conducting the online test.





IMPLEMENTATION SCREENSHOTS

Output page: The Output Page displays the final score immediately after exam submission. It shows the total marks obtained and highlights wrong or unattempted questions along with correct answers. This provides instant feedback to students and improves transparency in evaluation. It ensures fast, accurate, and automated result processing.

Exam Completed! Your Score: 15 / 20

Question 2: Wrong ❌
Your Answer: 6
Correct Answer: 8

Question 7: Wrong ❌
Your Answer: 14
Correct Answer: 12

Question 10: Wrong ❌
Your Answer: Planet
Correct Answer: Star

Question 13: Wrong ❌
Your Answer: 1945
Correct Answer: 1947

Question 14: Wrong ❌
Your Answer: 01
Correct Answer: 10

ADVANTAGES & LIMITATIONS

Advantages:

- Instant Result Generation: Provides real-time updates on stock levels and inventory movements automatically.
- Reduced Human Error: Minimizes manual entry mistakes in stock counting and evaluation through automation.
- Efficiency: Saves significant time and physical paperwork compared to traditional ledger-based tracking.
- User-Friendly Interface: Typically designed to be easy to use for warehouse and managers alike.

Limitations:

- Hardware Dependency: Requires a computer, tablet, or specialized scanning devices (like barcode or RFID readers) to access and update the system.
- Data Persistence: Depending on the version or tier of the software, basic systems might not offer long-term historical data storage or advanced backups.
- Security Concerns: May have limited security features compared to enterprise-grade ERP systems, making it vulnerable to unauthorized access if not properly managed.
- Connectivity Requirements: Often depends on a stable internet or local network connection for real-time synchronization across different locations.

Thank you